

Exercise sheet 8 for Numerical Optimization

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Exercise 1 (Trust region methods, 8+7+1+1 points)

The last approach for unconstrained optimization we get to know are *trust region methods*. The underlying idea is to approximate the objective function by a quadratic function and then minimize the quadratic function. Suppose the k -th iterate x_k , then Taylor's expansion yields the quadratic approximation

$$f(x_k + p) \approx q_k(p) := f(x_k) + \nabla f(x_k)^T p + \frac{1}{2} p^T H_f(x_k) p . \quad (1)$$

However, this approximation can only be reasonable in a neighbourhood around x_k . Therefore, we define a *trust region*

$$R_k := \{p \in \mathbb{R}^n : \|p\|_2 \leq \Delta_k\}$$

with radius $\Delta_k > 0$ in which we assume the approximation to be good enough. In order to ensure reasonability of the solution, we only minimize (1) inside this trust region. The resulting *trust region subproblem* is

$$\text{find } p_k \in R_k \quad \text{such that} \quad q_k(p_k) = \min_{p \in R_k} q_k(p) . \quad (2)$$

The next iterate is the optimal step inside the trust region,

$$x_{k+1} = x_k + p_k .$$

How to determine the trust region radii Δ_k ? Suppose p_k is the solution of (2). The reduction p_k means for the quadratic approximation is

$$\text{rq}_k = q_k(0) - q_k(p_k) = f(x_k) - q_k(p_k) = - \left(\nabla f(x_k)^T p_k + \frac{1}{2} p_k^T H_f(x_k) p_k \right) ,$$

the reduction p_k means for the function f is

$$\text{rf}_k = f(x_k) - f(x_k + p_k)$$

and the ratio of both will be denoted as

$$\rho_k = \frac{\text{rf}_k}{\text{rq}_k} .$$

If ρ_k is close to zero, the reduction in the quadratic model is much bigger as the reduction in f . From that, we conclude that the quadratic model is a bad approximation of f at $x_k + p_k$ and that the trust region is too large. Therefore, we reject the step and restart at x_k with a trust region with smaller radius. If ρ is significantly bigger than zero, we accept the step. If ρ_k is even close to one and the step reaches the boundary of the trust region, we consider the quadratic approximation to be too good. That means that with a bigger radius we might still have gotten an acceptable approximation but a larger step. So we probably could have been more efficient. We still accept the step, but increase the trust region radius for the next iteration step.

How to solve the trust region subproblems (2)? Of course, one could try to solve the trust region subproblems (2) exactly. However, except for special cases, this is not trivial and would mean a lot of effort. Besides, we do not know methods to solve such kinds of problems yet (it is an optimization with respect to constraints).

Because of that, often not the trust region subproblem itself is solved (an optimization with respect to the whole of R_k). Instead, only a line search along the steepest descent $-\nabla f(x_k)$ is performed. So the actual subproblem becomes

$$\text{find } p_k \in R_k \cap \text{span}\{-\nabla f(x_k)\} \quad \text{such that} \quad q_k(p_k) = \min_{p \in R_k \cap \text{span}\{-\nabla f(x_k)\}} q_k(p) . \quad (3)$$

We hope that this is a good approximation to the original subproblem. The big advantage is that the solution of (3) can be computed explicitly, it is called *Cauchy point*.

a) Prove that

$$p_k = \begin{cases} -\frac{\Delta_k}{\|\nabla f(x_k)\|_2} \nabla f(x_k) & \text{if } \nabla f(x_k)^T H_f(x_k) \nabla f(x_k) \leq 0 \\ \max\left(-\frac{\|\nabla f(x_k)\|_2^2}{\nabla f(x_k)^T H_f(x_k) \nabla f(x_k)}, -\frac{\Delta_k}{\|\nabla f(x_k)\|_2}\right) \nabla f(x_k) & \text{if } \nabla f(x_k)^T H_f(x_k) \nabla f(x_k) > 0 \end{cases}$$

solves (3).

The resulting algorithm is:

Algorithm 1: Trust region method with Cauchy step

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1 Input:  $f \in C(\mathbb{R}^n, \mathbb{R})^2$ , starting point  $x_0 \in \mathbb{R}^n$ , termination tolerance  $tol > 0$ , initial
   trust region radius  $\Delta_0 > 0$ , parameters  $0 < \eta_1 < \eta_2 < 1$  and  $0 < \nu_1 < 1 < \nu_2$ 
2 Output: approximation  $x_k$  of a stationary point
3  $k \leftarrow 0$ 
4 while  $\|\nabla f(x_k)\|_2 \geq tol$  do
5   // determine step
6   if  $\nabla f(x_k)^T H_f(x_k) \nabla f(x_k) \leq 0$  then
7      $p_k \leftarrow -\frac{\Delta_k}{\|\nabla f(x_k)\|_2} \nabla f(x_k)$ 
8   else
9      $p_k \leftarrow \max\left(-\frac{\|\nabla f(x_k)\|_2^2}{\nabla f(x_k)^T H_f(x_k) \nabla f(x_k)}, -\frac{\Delta_k}{\|\nabla f(x_k)\|_2}\right) \nabla f(x_k)$ 
10  // accept or reject step
11   $\rho_k \leftarrow \frac{f(x_k) - f(x_k + p_k)}{-\nabla f(x_k)^T p_k - \frac{1}{2} p_k^T H_f(x_k) p_k}$ 
12  if  $\rho_k > \eta_1$  then
13     $x_{k+1} \leftarrow x_k + p_k$ 
14  else
15     $x_{k+1} \leftarrow x_k$ 
16  // determine new trust region radius
17  if  $\rho_k \leq \eta_1$  then
18     $\Delta_{k+1} \leftarrow \nu_1 \Delta_k$ 
19  else if  $\rho_k \geq \eta_2 \wedge |\Delta_k - \|p_k\|_2| = 0$  then
20     $\Delta_{k+1} \leftarrow \nu_2 \Delta_k$ 
21  else
22     $\Delta_{k+1} \leftarrow \Delta_k$ 
23   $k \leftarrow k + 1$ 

```

b) Create a MATLAB programme `trust_region_cauchy.m` that implements algorithm 1. You should pass to it

- function handles `f`, `grad_f`, `H_f`,
- a starting point x_0 ,
- a termination tolerance `tol` and a maximum number of iteration steps `max_it`,
- an initial trust region radius Δ_0
- and parameters $\eta_1, \eta_2, \nu_1, \nu_2$.

The iteration is supposed to terminate if the norm of the gradient is smaller than `tol` or if the maximum number of iteration steps is reached. Let the programme return two matrices,

- an $(n \times (\text{number of iteration steps} + 1))$ -dimensional matrix containing all iterates
- and an $(n + 1)$ -dimensional vector containing the trust region radii of all iteration steps.

c) The MATLAB programme `test_trust_region_rosenbrock.m` tests your implementation on Rosenbrock's function. Execute the programme and reflect on the results.

d) Do the same for `test_trust_region_himmelblau.m`.

Exercise 2 (Cones, 2+2+3+4 points)

The next major chapter will be optimization under constraints. To do this, we first need to deal with cones. A set $K \subset \mathbb{R}^n$ is called cone, if for every $x \in K$ and $\alpha \geq 0$ it is $\alpha x \in K$.

- a) First, a simple example to illustrate the concept. Consider

$$K = \left\{ \alpha x \mid \alpha \geq 0, x \in S_1, S_1 = \left\{ \beta \begin{pmatrix} -1 \\ 1 \end{pmatrix} + (1 - \beta) \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \beta \in [0, 1] \right\} \right\} \subset \mathbb{R}^2 .$$

Draw S_1 and K .

- b) Find a set $S_2 \subset \mathbb{R}^2$ with $S_2 \neq \mathbb{R}^2$ such that the cone

$$K = \left\{ \alpha x \mid \alpha \geq 0, x \in S_2 \right\} \subset \mathbb{R}^2$$

equals $\mathbb{R}^2$.

- c) Cones do not need to be convex. Find an example of a non-convex cone.

- d) Prove that $K \subset \mathbb{R}^n$ is a convex cone if and only if it holds that

$$\forall x, y \in K \quad \forall \alpha, \beta \geq 0 \quad \text{it is} \quad \alpha x + \beta y \in K .$$

Exercise 3 (Farkas' Lemma (and hyperplanes and separability), 3+3+7+2+7 points)

Farkas' Lemma is a central tool for deriving optimality conditions in optimization with constraints. We want to get to know it and prove it.

To do this, we will first take a look at hyperplanes. A hyperplane $H \subset \mathbb{R}^n$ is a set

$$H = \left\{ x \in \mathbb{R}^n \mid x = p + \sum_{i=1}^{n-1} s_i u_i, \ s_1, \dots, s_{n-1} \in \mathbb{R} \right\}$$

with $p \in \mathbb{R}^n$ and linearly independent directions $u_1, \dots, u_{n-1} \in \mathbb{R}^n$. The direction vectors form an $(n-1)$ -dimensional affine coordinate system.

a) For $n = 3$, consider the hyperplane

$$H = \left\{ x \in \mathbb{R}^3 \mid x = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + s_1 \begin{pmatrix} \frac{1}{2} \\ 0 \\ \frac{1}{2} \end{pmatrix} + s_2 \begin{pmatrix} 0 \\ -\frac{1}{2} \\ -\frac{1}{2} \end{pmatrix} \right\} . \quad (4)$$

Visualize the hyperplane drawing it by hand, using MATLAB or any other tool. Check (calculating by hand) if the points

$$q_1 = \begin{pmatrix} \frac{3}{2} \\ -1 \\ -\frac{1}{2} \end{pmatrix} \quad \text{and} \quad q_2 = \begin{pmatrix} \frac{5}{2} \\ -1 \\ -\frac{1}{2} \end{pmatrix}$$

lie on H .

Every hyperplane can be represented in normal form

$$H = \{x \in \mathbb{R}^n \mid v^T x = v^T p\} ,$$

where $v \in \mathbb{R}^n \setminus \{0\}$ is a normal vector of the hyperplane.

b) Express the hyperplane (4) in normal form and add the normal vector v to your visualization from exercise a). Note that the normal form is unique up to the length of the normal vector v .

Check again with the normal form, if q_1 and q_2 lie in H .

A hyperplane divides $\mathbb{R}^n$ into two half-spaces, namely

$$\{x \in \mathbb{R}^n \mid v^T x \leq v^T p\} \quad \text{and} \quad \{x \in \mathbb{R}^n \mid v^T x \geq v^T p\} .$$

The intersection of the interiors of both half-spaces is empty, they only meet on the hyperplane. So, every $x \in \mathbb{R}^n$ with $v^T x < v^T p$ lies on one side of the hyperplane and every $x \in \mathbb{R}^n$ with $v^T x > v^T p$ on the other side of the hyperplane.

The **separation theorem** yields a geometrical statement about convex sets. It can be formulated in many variants, we are interested in the following strong variant:

Consider $K \subset \mathbb{R}^n$ convex with $K \neq \emptyset$. Also consider $y \in \mathbb{R}^n$ with $y \notin \overline{K}$. Then the two sets $\{y\}$ and K are separable, meaning that there exists a $v \in \mathbb{R}^n \setminus \{0\}$ such that $v^T y < v^T x \ \forall x \in K$.

We now ask what happens when K is a non-empty, convex and closed cone.

- c) Let $K \subset \mathbb{R}^n$, $K \neq \emptyset$, be a convex and closed cone. Prove using the separation theorem that for every $y \notin K$ there exists a $v \in \mathbb{R}^n \setminus \{\mathbf{0}\}$ such that for all $x \in K$ it holds that $v^T y < 0 \leq v^T x$.
- d) What does the statement in exercise c) mean with regard to non-empty, convex and closed cones, vectors $y \in \mathbb{R}^n$ and the existence of hyperplanes and the position of y and K in the half-spaces?

Farkas' Lemma states the following: Consider a matrix $B \in \mathbb{R}^{k \times n}$ and a vector $d \in \mathbb{R}^k$. Then exactly one of the two statements is true (order relations applied to vectors are meant elementwise):

- The problem

$$Bx = d, \quad x \geq 0$$

has a solution $x \in \mathbb{R}^n$.

- The problem

$$B^T \lambda \geq 0, \quad \lambda^T d < 0$$

has a solution $\lambda \in \mathbb{R}^k$.

- e) Prove Farkas' Lemma using the statement from exercise c).

Hint: First, show that $K = \{Bx \mid x \geq 0, x \in \mathbb{R}^n\} \subset \mathbb{R}^k$ is a non-empty, convex and closed cone. Second, consider the two cases $d \in K$ and $d \notin K$.

Farkas' Lemma looks quite abstract. But it implies, for example, the so-called KKT conditions. These are necessary first-order optimality conditions for optimization under constraints. In other words, they are the generalization of $\nabla f(x) = \mathbf{0}$ when there are constraints. We will deal with them soon.